SERVER AND NETWORK PREPARATION

Server and network preparation is a critical phase in IT infrastructure deployment. It involves configuring hardware, software, and communication systems to ensure reliable, secure, and efficient data exchange across networks. Proper preparation minimizes downtime, enhances performance, and supports scalability.

A server is a computer system that provides resources, data, or services to other computers (clients) over a network.

Types of Servers

1. File Servers

File servers are specialized systems responsible for **centralized storage, management, and sharing of files** across a network.

Functions

- Store documents, images, videos, and organizational data
- Provide controlled access to users
- Support file sharing and collaboration

Key Features

- Access control (permissions: read, write, execute)
- Backup and recovery mechanisms
- File synchronization

Examples of Use

- University systems storing student records
- Organizational shared drives

2. Web Servers

Web servers host and deliver web content (websites and applications) to users over the internet or intranet.

Functions

- Process HTTP/HTTPS requests from clients (browsers)
- Serve static content (HTML, CSS, images)
- Support dynamic content via server-side scripting

Common Web Server Software

- Apache HTTP Server
- Nginx
- Microsoft IIS

Real-World Application

- Hosting e-commerce platforms
- Running institutional websites

3. Database Servers

Database servers manage structured data using Database Management Systems (DBMS).

Functions

- Store, retrieve, and manage data
- Execute queries (SQL)
- Ensure data integrity and security

Key Features

- Transaction management
- Data consistency (ACID properties)
- Backup and replication

Examples

- MySQL
- PostgreSQL
- Oracle Database
- Microsoft SQL Server

4. Application Servers

Application servers provide an environment for running business logic and enterprise applications.

Functions

- Execute application code
- Process user requests
- Connect web servers to databases

Examples of Application Servers

- Apache Tomcat
- JBoss
- WebLogic

Use Cases

- Online banking systems
- University management systems

5. Mail Servers

Mail servers handle sending, receiving, and storing emails.

Protocols Used

- SMTP (sending emails)
- POP3 (retrieving emails)
- IMAP (synchronizing emails across devices)

Functions

- Manage user mailboxes
- Filter spam
- Ensure secure communication

Examples

- Microsoft Exchange Server
- Postfix
- Sendmail

Hardware Preparation

Key Components

1. CPU (Processor)

- Acts as the "brain" of the server
- Determines how many tasks can be processed simultaneously
- Multi-core processors improve performance in multi-user environments

2. RAM (Random Access Memory)

- Stores temporary data for quick access
- More RAM allows better multitasking and faster response times
- Critical for database and application servers

3. Storage

- **HDD (Hard Disk Drive):** Cheaper but slower

- **SSD (Solid State Drive):** Faster, more reliable, preferred for servers

Storage Considerations

- Capacity requirements
- Speed (read/write performance)
- Redundancy

4. Network Interface Card (NIC)

- Connects the server to the network
- Determines data transfer speed (1Gbps, 10Gbps, etc.)
- Supports multiple connections in high-performance systems

Best Practices

1. Redundant Power Supplies

- Prevent downtime in case of power failure
- Common in enterprise servers

2. RAID Configuration

RAID (Redundant Array of Independent Disks) improves performance and fault tolerance.

Common RAID Levels

- RAID 0: Performance (no redundancy)
- RAID 1: Mirroring (high reliability)
- RAID 5: Balance of performance and redundancy

3. Cooling Systems

- Prevent overheating
- Includes fans, air conditioning, and liquid cooling

- Ensures system longevity and stability

Software Installation

Operating Systems

1. Windows Server

- User-friendly interface
- Strong integration with enterprise tools (Active Directory)
- Suitable for organizations using Microsoft ecosystems

2. Linux Distributions

- Open-source and cost-effective
- Highly customizable
- Popular options:
    - Ubuntu Server
    - CentOS / Rocky Linux
    - Red Hat Enterprise Linux (RHEL)

Installation Steps

Step 1: Install Operating System

- Boot from installation media (USB/DVD)
- Partition disks
- Configure initial settings

Step 2: Apply Updates and Patches

- Fix security vulnerabilities
- Improve performance and stability

Step 3: Install Server Roles

Server roles define the server's function.

Examples

- DNS Server
- DHCP Server
- Web Server (Apache, IIS)
- File Server

2.4 Server Configuration

Basic Configurations

1. Assign Static IP Address

- Ensures the server always has the same IP
- Essential for reliable access

2. Configure Hostname

- Unique name identifying the server on the network
- Example: webserver01.company.local

3. Domain Membership

- Join server to a domain (e.g., Active Directory)
- Enables centralized management and authentication

Service Configuration

1. DNS Server

- Translates domain names into IP addresses
- Example: www.example.com → 192.168.1.10

2. DHCP Server

- Automatically assigns IP addresses to devices
- Reduces manual configuration errors

3. Web Server Configuration

- Install and configure:
  - Apache
  - Nginx
  - IIS
- Set document root and security settings

Security Preparation

1. Firewalls

- Control incoming and outgoing traffic
- Prevent unauthorized access
- Examples: hardware and software firewalls

2. Encryption (SSL/TLS)

- Protects data during transmission
- Used in HTTPS for secure web communication

3. User Authentication and Access Control

- Use strong passwords and multi-factor authentication (MFA)
- Implement role-based access control (RBAC)
- Limit user privileges

4. Regular Patching and Updates

- Protect against vulnerabilities and cyber threats

- Maintain system integrity

Additional Security Measures

- Intrusion Detection Systems (IDS)
- Antivirus/anti-malware software
- Backup and disaster recovery plans

3. Network Installation and Configuration

3.1 Network Planning

Before installation:

Considerations

- Network size and topology
- Type of network (LAN, WAN, WLAN)
- Bandwidth requirements
- Security requirements

Network Topologies

- Star topology
- Bus topology
- Ring topology
- Mesh topology

Network Hardware Installation

Devices

- Routers
- Switches
- Access points
- Modems

Cabling

- Ethernet cables (Cat5e, Cat6, Cat7)
- Fiber optic cables

IP Addressing and Configuration

IPv4 Address Structure

- Example: 192.168.1.1
- Includes network and host portions

Subnetting

Divides networks into smaller segments for efficiency.

Core Configuration Tasks

- Assign IP addresses (static/dynamic)
- Configure subnet mask and gateway
- Set up DNS servers
- Configure routing protocols

Wireless Network Configuration

- Set SSID
- Enable WPA3/WPA2 security

- Configure access control
- Optimize signal coverage

Troubleshooting Network Systems

1 Common Network Problems

- No connectivity
- Slow network performance
- IP conflicts
- DNS resolution failure
- Packet loss

2 Troubleshooting Methodology

1. Identify the Problem: This step involves clearly understanding what the issue is. The technician gathers information by asking users, observing symptoms, and checking system logs or error messages. The goal is to determine the exact nature and scope of the problem, including when it started and who or what is affected.

2. Establish a Theory of Probable Cause: After identifying the problem, possible causes are suggested based on the symptoms. This involves logical thinking, past experience, and technical knowledge. The technician considers the most likely reasons for the issue, starting with the simplest or most common causes.

3. Test the Theory: In this step, the suspected cause is tested to confirm whether it is correct. Diagnostic tools and methods are used to verify the theory. If the test proves the theory wrong, another possible cause is considered and tested until the real cause is identified.

4. Implement the Solution: Once the actual cause of the problem is confirmed, the appropriate solution is applied. This may involve fixing configurations, replacing faulty hardware, restarting systems, or updating software. Care is taken to ensure the solution does not create new problems.

5. Verify Functionality: After implementing the solution, the system is tested to ensure that the problem has been fully resolved. The technician confirms that all affected services are working correctly and that users can operate normally without issues.

6. Document Findings: Finally, the entire process is recorded, including the problem, its cause, the solution applied, and the outcome. Documentation helps in future troubleshooting, knowledge sharing, and improving system maintenance practices.

Diagnostic Tools

- **Ping (Packet Internet Groper): Tests Connectivity**: Ping is used to check whether a device can communicate with another device over a network. It sends small data packets to a target IP address or domain and waits for a response. If a reply is received, it confirms that the connection is active; if not, there may be a network issue such as disconnection or blocking.
- **Traceroute: Tracks Packet Path**: Traceroute identifies the path that data packets take from the source to the destination. It shows each intermediate device (router) along the route and the time taken to reach each point. This helps detect where delays or failures occur within the network.
- **ipconfig / ifconfig: Displays IP Configuration**: These commands display the network configuration of a device.

- **ipconfig** is used in Windows systems
- **ifconfig** is used in Linux/Unix systems

They show details such as IP address, subnet mask, and default gateway, helping to diagnose configuration issues like incorrect IP settings.

- **Netstat (Network Statistics): Shows Network Connections**: Netstat displays active network connections, listening ports, and network statistics. It helps identify which

services are running, detect unauthorized connections, and troubleshoot issues related to ports and communication.

- **Nslookup: Tests DNS Resolution**: Nslookup is used to check how domain names are translated into IP addresses by DNS servers. It helps diagnose DNS-related issues, such as when a website cannot be accessed due to incorrect or failed name resolution.

4 Troubleshooting Layers

- Physical layer: cables, hardware
- Data link: MAC addresses, switches
- Network layer: IP routing
- Transport layer: ports and protocols

Common Solutions

- Restart devices
- Reconfigure IP settings
- Replace faulty hardware
- Update drivers and firmware

Network Optimization Techniques

1 Importance of Optimization

Optimization improves network speed, efficiency, and reliability while reducing latency and congestion.

2 Bandwidth Management

- Allocate bandwidth based on priority
- Use Quality of Service (QoS)

3 Load Balancing

Distributes network traffic across multiple servers to prevent overload.

4 Network Segmentation

- Use VLANs to divide networks
- Improves security and performance

5 Caching and Compression

- Reduce data transmission size
- Improve response time

6 Hardware Optimization

- Upgrade routers and switches
- Use high-speed NICs
- Implement fiber optics

7 Monitoring and Performance Analysis

- Use monitoring tools (e.g., Nagios, Zabbix)
- Analyze traffic patterns
- Detect bottlenecks

8 Security Optimization

- Intrusion Detection Systems (IDS)
- Intrusion Prevention Systems (IPS)
- Regular vulnerability assessments